Interface agents for an automated biomanufacturing DBTL pipeline

Case study: (2S)-pinocembrin in E. coli · SYNBIOCHEM pipeline, Carbonell et al., Communications Biology (2018)

- Reported DBTL stages and tools
- Proposed interface agents

4 transition agents · 1 coordination layer

01 Design

Select pathway, enzymes and parts library

TECHNIQUES · EQUIPMENT · SOFTWARE

- RetroPath · Selenzyme selection
- PartsGenie · RBS + codon design
- SBC-DoE · DominoGenie worklists

D→B | Design-to-Build

- Expand DoE library into part lists
- Flag risky junctions · bridging oligos
- Register parts in ICE · emit worklists

HANDOFF: construct library + LCR recipes

02 Build

Assemble and verify pathway plasmids

TECHNIQUES · EQUIPMENT · SOFTWARE

- DNA synthesis · PCR part prep
- Ligase cycling · Hamilton STAR
- Digest + CE · Sanger verification

B→T | Build-to-Test

- Reconcile verified vs failed constructs
- Map plasmid IDs to wells and runs
- Release only sequence-perfect clones

HANDOFF: verified strains + plate map

03 Test

Cultivate strains and quantify titers

TECHNIQUES · EQUIPMENT · SOFTWARE

- Chassis transformation · 96-DW growth
- Automated extraction · UPLC-MS/MS
- MRM quantification · R processing

T→L | Test-to-Learn

- Gate peaks by LOD / LOQ and calibration
- Normalize titers across replicate clones
- Join titers to design factor table

HANDOFF: curated titer-design dataset

04 Learn

Infer design rules from titer data

TECHNIQUES · EQUIPMENT · SOFTWARE

- Least-squares factor analysis (JMP)
- Statistical machine learning models
- Part-effect ranking · design rules

L→D | Learn-to-Design

- Rank factor effects and uncertainty
- Turn effects into library constraints
- Decide DoE vs full-factorial next round

HANDOFF: focused library specification

Coordinator

cross-stage oversight

- FAIR provenance (ICE / SBOL)
- Handoff validation
- Exception handling · failed builds
- Human approval gates

Optimised → process development

Chassis and media screening, genome integration, scale-up

Solid arrows: DBTL flow through interface agents · Dashed lines: coordination and oversight · Green: exit to scale-up

Proposed architecture · not implemented in the original study